PSYCHOLOGY

Chapter 2:

The Biological Approach

Welcome to AnithUncommon's **Sample Resources!**

We are assuming you are reading this as a student. So what will **YOU** expect from this sample?

In Psychology, The Biological Approach, 2.1-2.4…

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Sharia Shifa to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Psychology, join our nonprofit!

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This sample is made thanks to, Sharia Shifa

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to…

- **Understand** the biological assumption that behavior is a product of physiological processes.
- **State** the relationship between brain structures (like the amygdala) and emotional memory.
- **Investigate** the physiological correlates of sleep, dreaming, and emotional arousal.

Reflective Questions to Consider:

The biological approach in psychology explains behavior by looking at the brain, hormones, and body processes. It focuses on how physical changes inside the body can affect thoughts and emotions.

So, think about these:

- What part of the brain is linked to emotions like fear or memory?
- How can hormones affect how a person feels or behaves?
- Do you think all human behavior can be explained just by biology? Why or why not?

Unit 2: The Biological Approach

Tip!
In the biological approach, don't try to memorize brain parts, sleep stages, and hormones separately. Instead, always ask: "What does this do to behavior?" For example, the amygdala links to emotions like fear, REM sleep links to dreaming, and hormones like adrenaline affect arousal. If you connect every biological concept to its effect on behavior, the whole chapter becomes easier to organize in your mind instead of feeling like random facts.

Mnemonic (to remember the BIG idea):
"BRAIN = Behavior, Reactions, Arousal, Inside Nervous-system"

Activities in Class:

- Lesson Breakdown
- Brain Map Drawing: Label the amygdala and explain its function.
- Sleep Log Analysis: Use the Dement & Kleitman findings to hypothesize why we remember certain dreams over others.

2.1 The Emotional Brain (Canli et al., 2000)

Amygdala and Emotion

The amygdala is a small structure in the brain that plays a major role in processing emotions, especially fear and emotional intensity. It acts like an "emotional detector," helping the brain quickly respond to important or threatening situations.

It also strengthens emotional memories. This means events that are emotionally intense (like fear or happiness) are more likely to be remembered than neutral events.

In Canli et al. (2000), participants viewed emotional images while their brains were scanned. The results showed that when the amygdala was more active, participants were more likely to remember those images later.

Mentor's Notes

The biological approach is based on one key idea: behavior comes from inside the body. If you understand this, everything else becomes easier to connect. The amygdala shows how brain structures influence emotion, sleep studies show how brain activity changes during rest, and Schachter & Singer show how the body and

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- This suggests that the amygdala does not just process emotion — it also helps store emotional memories

Neuroimaging (fMRI)

Functional Magnetic Resonance Imaging (fMRI) is a technique used to observe brain activity in real time.

It works by detecting changes in blood flow. When a part of the brain becomes active, it uses more oxygen, and fMRI tracks this change to identify which areas are involved.

- This allows psychologists to “see” which parts of the brain are working during tasks like thinking, remembering, or feeling emotions.

2.2 The Physiology of Sleep (Dement & Kleitman, 1957)

REM Sleep and Dreaming

REM (Rapid Eye Movement) sleep is a stage of sleep where most vivid dreaming occurs. It is called “rapid eye movement” because the eyes move quickly under the eyelids during this stage.

Even though the brain is highly active during REM sleep, the body remains mostly still. This prevents us from physically acting out our dreams.

Dement & Kleitman found that when participants were woken up during REM sleep, they were more likely to report detailed dreams compared to when they were woken during non-REM sleep.

- This shows a strong connection between REM sleep and dreaming.

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mind work together to create emotion. Notice the pattern: brain → body → behavior

Each study in this chapter focuses on a different level of biology. *Canli et al*. focuses on the brain and memory, *Dement and Kleitman* focuses on sleep cycles and brain activity, and *Schachter and Singer* combines biology with cognition. Together, they show that behavior is not random but linked to measurable internal processes.

Tip!
Try to think of the brain like a control room:

Amygdala = alarm system (emotion detector)

REM sleep = night mode recording

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EEG and EOG EEG (Electroencephalogram) records electrical activity in the brain, allowing researchers to identify different stages of sleep. EOG (Electrooculogram) measures eye movements, which helps detect when REM sleep is occurring Together, these tools make sleep research more objective, instead of relying only on what people say they experienced. **2.3 Two-Factor Theory of Emotion (Schachter & Singer, 1962)** **<u>Arousal + Cognition</u>** ✔ Emotion is created by two factors: physiological **arousal** and **cognitive** interpretation. • First, the body experiences arousal (e.g. increased heart rate). • Then, the brain labels this arousal based on the situation (e.g. fear, excitement, anger). This means the body reacts first, and then the brain decides what that feeling means. Example: Fast heartbeat + dangerous situation = fear Fast heartbeat + exciting event = excitement **<u>Interpretation of Emotion</u>** ✔ The same physical arousal can lead to different emotions depending on how the situation is interpreted. ✔ This shows that emotion is not purely biological, but also influenced by thinking and context.	**system (dreaming happens here)** **Arousal = body "energy signal"** **Cognition = brain's interpretation of what the signal means** **Tip!** **The key pattern to notice: biology explains the *hardware*, not always the *software*. The brain and hormones create the base reaction, but interpretation (thinking) can change the final emotion. That's why some studies in this chapter are purely biological (Canli, Dement), while others mix biology with cognition (Schachter & Singer).**

So, emotions are a combination of **body + mind**, not just one. **2.4 Issues, Debates and Approaches** **Nature vs Nurture** The biological approach mainly supports **nature**, meaning behavior is influenced by genes, brain structures, and hormones. However, Schachter & Singer show that cognition and environment also influence emotion, meaning nurture also plays a role. • This means behavior is usually influenced by both biology and experience. **Reductionism** Reductionism means explaining complex behavior by breaking it down into simpler biological parts, such as brain activity or hormones. The biological approach is considered reductionist because it focuses mainly on physical processes inside the body. While this makes behavior easier to study scientifically, it may ignore thoughts, emotions, and social influences. **Evaluation Idea** Biological explanations are strong because they are **scientific, measurable, and objective** (e.g., brain scans, EEG results). However, they can be **limited** because they may ignore personal experiences, thoughts, and environmental factors. So, while biology explains *how* the body reacts, it may not fully explain *why* people feel or behave in certain ways	

Mentors' Sources:

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Canli et al. (2000) – Amygdala and emotional memory

Dement & Kleitman (1957) – Sleep and dreaming

Schachter & Singer (1962) – Two-factor theory of emotion

Stanley Schachter – Developed the two-factor theory of emotion

Matthew Lieberman – Work on brain and emotion (helps support biological explanations)

Joseph LeDoux – Research on the amygdala and fear